

딥러닝 최종보고서

Generation of Personalized Avatar Stickers based on Deep Learning

2022203502 CHOI VADIM (최바딤)

Abstract: This paper presents a deep learning-based method for generating personalized cartoon avatar stickers using a user's facial image and an emoji representing the target emotion. The motivation stems from the growing demand for expressive digital avatars, but the development of such a system is often limited by data availability. I detail a two-stage pipeline to address this challenge. First, a high-quality training dataset is created by leveraging a pre-existing paired dataset and enhancing it with a modern diffusion model to programmatically edit emotions onto the stickers. Second, I train a custom dual-encoder U-Net model on this new dataset for the final, real-time inference task. The face encoder, implemented as the U-Net's contracting path, is trained from scratch to extract identity-preserving features specifically tailored to the sticker domain. A parallel lightweight CNN encodes emotion signals from an emoji image. These feature maps are fused in the bottleneck and processed by the U-Net's expansive path, which uses skip-connections to reconstruct the final 128x128 RGBA sticker. This approach validates a scalable proof of concept, showing promise in identity preservation and emotional expression within the project's resource constraints.

Keywords: Deep Learning, U-Net, Generative Models, Image-to-Image Translation, Conditional Generation

1. Introduction

With the rise of digital communication, personalized visual content such as stickers has become increasingly important. Stickers that reflect a user's real appearance and emotions are in high demand, yet their creation is often limited to static templates or requires labor-intensive customization.

I propose an automated solution based on deep learning. I developed a system that generates a cartoon-style sticker from a user's photo and an emoji. I specifically chose emojis as the secondary input modality because users are highly familiar with them and intuitively use them to express emotions, making them a more universal tool than text labels.

The primary challenge in this domain is the absence of high-quality, paired datasets. Initial research revealed that creating such a dataset from scratch is a significant undertaking. This led me to develop a two-stage approach that uses modern diffusion models for offline data preparation, which then powers my custom, lightweight U-Net model for online generation. This report details the data synthesis pipeline and the final model architecture.

2. Proposed Method

My final pipeline consists of two independent components: offline dataset generation and the real-time sticker generation model.

2.1. Dataset Generation via Inpainting

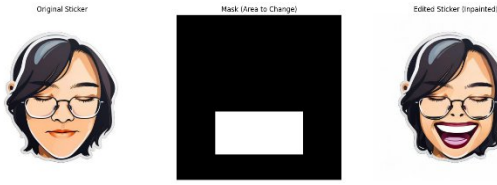
I address the data scarcity problem by modifying an existing dataset using modern tools.

2.1.1 Base Data

I use the public Hugging Face dataset `misshimichka/face_stickers_cleared` as my foundation. It provides pre-existing pairs of (`real_face`, `neutral_sticker`).

2.1.2 Emotion Editing

To add emotions to the neutral stickers, I employ the `stabilityai/stable-diffusion-2-inpainting` diffusion model from the `diffusers` library. As shown in [Fig. 1], the process involves creating a mask over the facial expression area and using a targeted text prompt (e.g., "make the person smile") to repaint only that region.



[Fig. 1] The emotion editing process using a Stable Diffusion Inpainting pipeline.

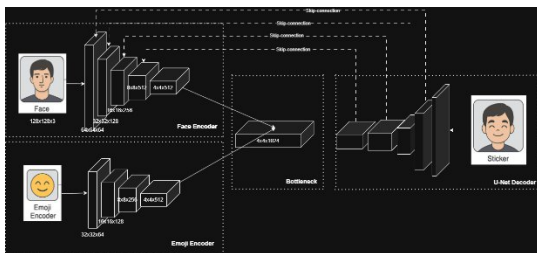
2.2. Final Model Architecture

For fast, real-time sticker generation, I designed and implemented a custom U-Net model with a dual-encoder architecture, which is trained from scratch. The architecture is detailed below and illustrated in [Fig. 2].

Face Encoder: The face encoder is implemented as the contracting path of the U-Net. It consists of a series of five convolutional blocks (a Conv2D layer with a stride of 2, followed by BatchNormalization and a ReLU activation) that progressively downsample the input image from 128x128 to a 4x4 bottleneck feature map. Unlike approaches that use pre-trained backbones, this encoder is trained from scratch, allowing it to learn feature representations specifically tailored to the cartoon sticker domain. The outputs from each block are preserved for use as skip-connections.

Emoji Encoder: A parallel, lightweight CNN with a similar downsampling structure processes the 64x64 emoji image. It converts the emoji into a 4x4 feature map with a channel depth matching the face encoder's bottleneck, capturing its emotional and stylistic cues.

Generator (Decoder): The final feature maps from both encoders are concatenated along the channel axis to form a fused bottleneck tensor. This tensor is then fed into the expansive path (decoder), which consists of a series of transposed convolutional layers (Conv2DTranspose). At each upsampling stage, the corresponding skip-connection from the face encoder is concatenated, which is crucial for allowing the model to recover fine-grained details and preserve the user's identity. The final layer is a convolutional layer with a sigmoid activation function that produces the 128x128 RGBA sticker.



[Fig. 2] The architecture of the final dual-encoder U-Net sticker generation model.

2.3. Justification for the Two-Stage Approach

A valid question arises: why not use a powerful diffusion model like Stable Diffusion directly to generate stickers on user demand, instead of training a separate model? The answer lies in the practical requirements for interactive applications: speed, cost, and controllability. My two-stage approach allows me to "distill" the power of a large, slow, and expensive model (used for offline data preparation) into a small, fast, and efficient model (my U-Net) designed for online use by the end-user. Since this model is intended for use within messaging apps, it must be lightweight and responsive so that users do not have to wait—stickers should be generated automatically based on the user's photo in real-time.

Moreover, since this model is tailored specifically for the sticker generation task, it can potentially outperform general-purpose models like Stable Diffusion in terms of relevance and quality for this domain. One of the major advantages is the ability to maintain a coherent and consistent visual style across all generated stickers. Unlike Stable Diffusion, which is trained for general-purpose image generation, my model is specialized to produce visually unified cartoon stickers that match the stylistic expectations of modern messaging platforms.

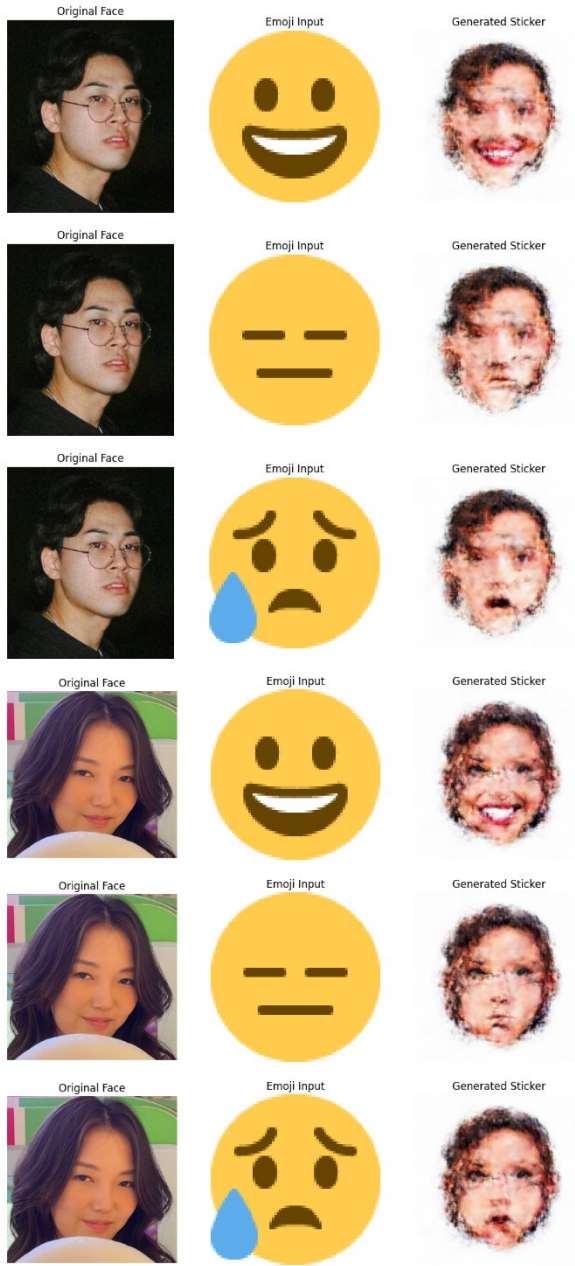
3. Experiments & Results

3.1. Training Details

The model was trained for 150 epochs using the Adam optimizer with a learning rate of 0.0002. The objective function was Mean Absolute Error (L1 loss). Due to hardware constraints, small batch sizes were used on a single GPU. I used a limited dataset of only 300 images—composed of 100 unique identities, each augmented with 3 distinct emotional variations. In the future, I aim to scale this to the full dataset of 700 identities, applying all emotion-relevant emoji variations to improve coverage and generalization.

3.2. Qualitative Results

The results, as expected given the extremely limited dataset, are relatively weak in terms of final sticker quality. However, even at this stage, the model shows encouraging signs of learning meaningful features: it can already distinguish between male and female faces, and responds to different emotional cues provided via the emoji input. As illustrated in [Fig. 3], this behavior is visible across several generated outputs.



[Fig. 3] Examples of generated stickers for two identities with various emoji inputs.

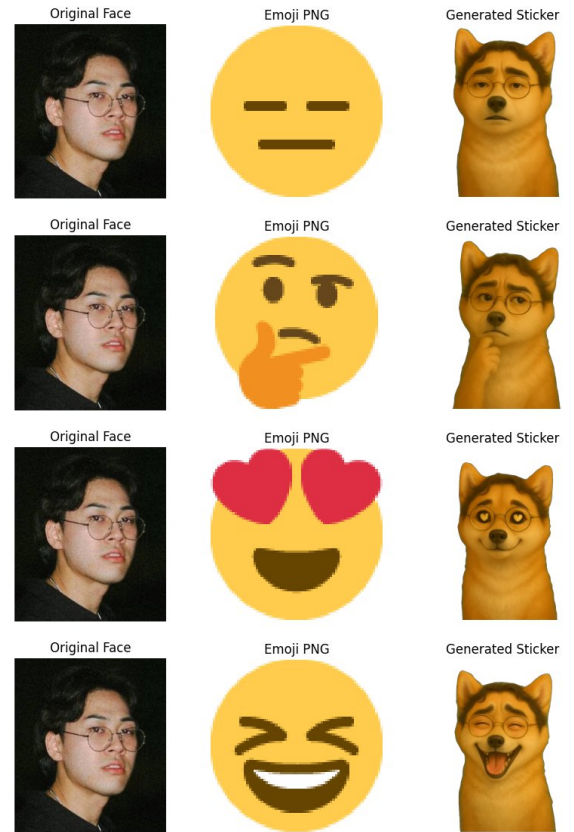
4. Conclusion and Future Work

In this project, I developed and trained a dual-encoder U-Net framework for generating personalized avatar stickers. The primary challenge of data scarcity was overcome by creating a synthetic dataset using a modern diffusion-based inpainting model. The final custom-trained model demonstrates a strong proof of concept, effectively preserving user identity while transferring emotional cues from emoji inputs. This two-stage

pipeline proves to be a viable strategy for creating specialized generative models under significant resource constraints.

In the future, I plan to significantly enhance the dataset by expanding it to cover all 700 identities and include a broader range of emotion-specific emojis. Additionally, I aim to incorporate advanced multimodal capabilities by leveraging models like ChatGPT-4.1, which would enable automated caption generation and richer emotional context. This approach could further enhance identity consistency and emotional nuance in the generated outputs.

To illustrate this vision, [Fig. 4] presents a reference example of the high-quality result I aim to achieve by scaling the dataset and using powerful generative and language models.



[Fig. 4] A target sample output generated using ChatGPT-4.1 (stylized and expressive sticker). (For future model goals)

5. Reference

- [1] P. Isola, et al., "Image-to-image translation with conditional adversarial networks," *CVPR*, 2017.
- [2] O. Ronneberger, P. Fischer, and T. Brox, "U-net:

Convolutional networks for biomedical image segmentation,"
MICCAI, 2015.

- [3] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer,
"High-resolution image synthesis with latent diffusion models,"
CVPR, 2022.
- [4] Hugging Face, "The diffusers library," [Online]. Available:
<https://github.com/huggingface/diffusers>
- [5] Hugging Face Dataset Viewer, "Face Stickers Dataset
(Cleared)," [Online]. Available:
https://huggingface.co/datasets/misshimichka/face_stickers_cleared/viewer/default/train?row=98&views%5B%5D=train
- [6] StabilityAI, "Stable Diffusion 2 Inpainting," [Online]. Available:
<https://huggingface.co/stabilityai/stable-diffusion-2-inpainting> Company, Country of Company, [Online], <http://www.com>.